

Math Foundations: Linear Algebra and Eigenproblems

Paired reading: 02_readings/00_prerequisites/Eigenproblems.fm.pdf **Focus area A**
(flagged weak area)

This is the math that UL3 (PCA/ICA) and SL6 (SVMs) are built on, plus every distance computation in SL4 (kNN) and UL1 (k-means). Close these gaps in Week 0, not when the unit exposes them. Where a derivation stalls you, log it in `math_gaps_log.md` and close it the same week.

1 Vectors, Norms, and the Inner Product

A **vector** $x \in \mathbb{R}^n$ is a point or direction in n -dimensional space. Write it as a column by default:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad x^\top = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}.$$

1.1 Inner Product: The One Operation Everything Reduces To

$$x \cdot y = x^\top y = \sum_i x_i y_i = \|x\| \|y\| \cos \theta.$$

The right-most identity is the geometric definition. It encodes three facts in one line:

- **Magnitude:** $x \cdot x = \|x\|^2$, so $\|x\| = \sqrt{x^\top x}$, the L_2 norm.
- **Angle:** $\cos \theta = \frac{x \cdot y}{\|x\| \|y\|}$ is the **cosine similarity**, used directly in NLP, IR, and as a similarity kernel.
- **Orthogonality:** $x \cdot y = 0 \iff x \perp y$. This is why SVMs care about inner products: the separating hyperplane is defined by $w \cdot x + b = 0$.

Properties drilled into you because every SVM/kernel derivation uses them:

$$\begin{aligned} \text{Symmetry:} \quad & x \cdot y = y \cdot x, \\ \text{Bilinearity:} \quad & (ax + bz) \cdot y = a(x \cdot y) + b(z \cdot y), \\ \text{Positivity:} \quad & x \cdot x \geq 0, \quad \text{with equality iff } x = 0. \end{aligned}$$

These three properties **define** an inner product abstractly. Anything satisfying them induces a valid geometry, which is exactly why a kernel $k(x, y)$ that secretly equals $\phi(x) \cdot \phi(y)$ in some other space still gives you a working SVM.

1.2 Norms Beyond L_2

Norm	Formula	Used in
L_1 (Manhattan)	$\ x\ _1 = \sum_i x_i $	Lasso, sparse regularization
L_2 (Euclidean)	$\ x\ _2 = \sqrt{\sum_i x_i^2}$	Ridge, SVM margin, kNN
L_∞ (max)	$\ x\ _\infty = \max_i x_i $	Adversarial robustness bounds
L_p (general)	$(\sum_i x_i ^p)^{1/p}$	–

The **Cauchy–Schwarz inequality**, used in every margin proof you will meet:

$$|x \cdot y| \leq \|x\| \|y\| \quad (\text{equality iff } x \text{ and } y \text{ are colinear}).$$

1.3 Projection: The Geometry of Best Linear Approximation

Given a unit vector u with $\|u\| = 1$, the projection of x onto the line spanned by u is

$$\text{proj}_u(x) = (x \cdot u)u.$$

The scalar $x \cdot u$ is the **coordinate** of x along u ; multiplying by u puts it back in $\mathbb{R}^n$. The residual $x - (x \cdot u)u$ is orthogonal to u . This single fact, applied to a set of orthonormal directions, is PCA’s reconstruction.

Generalized projection onto a subspace spanned by columns of U with $U^T U = I$:

$$\text{proj}_U(x) = U U^T x.$$

$U U^T$ is a **projection matrix**: symmetric ($P = P^T$) and idempotent ($P^2 = P$).

2 Matrices as Linear Transformations

A matrix $A \in \mathbb{R}^{m \times n}$ is a **linear map** $\mathbb{R}^n \rightarrow \mathbb{R}^m$. Read every matrix as “a machine that rotates, scales, shears, and projects space.” Linearity means

$$A(\alpha x + \beta y) = \alpha(Ax) + \beta(Ay).$$

That is it: no curvature, no offsets. Everything in this course assumes linearity; non-linear methods (kernels, neural networks) reach beyond it by applying a linear map after a non-linear transformation.

2.1 The Four Fundamental Subspaces

For any $A \in \mathbb{R}^{m \times n}$:

Subspace	Definition	Dimension
Column space $\text{Col}(A)$	$\{Ax : x \in \mathbb{R}^n\}$	$\text{rank}(A) = r$
Null space $\text{Null}(A)$	$\{x : Ax = 0\}$	$n - r$
Row space $\text{Row}(A) = \text{Col}(A^T)$	$\{A^T y : y \in \mathbb{R}^m\}$	r
Left null space $\text{Null}(A^T)$	$\{y : A^T y = 0\}$	$m - r$

$\text{Col}(A) \perp \text{Null}(A^T)$ in $\mathbb{R}^m$, and $\text{Row}(A) \perp \text{Null}(A)$ in $\mathbb{R}^n$. The **rank–nullity theorem** says $\dim \text{Col}(A) + \dim \text{Null}(A) = n$.

Why this matters: in linear regression $y \approx X\beta$, the closest you can ever get y to is its projection onto $\text{Col}(X)$. The residual lives in $\text{Null}(X^T)$, which is why the normal equations come out as $X^T(y - X\beta) = 0$.

2.2 Special Matrices You Must Recognize on Sight

- **Symmetric:** $A = A^T$. Covariance matrices, Gram matrices, and Hessians of scalar functions. Spectral theorem applies: real eigenvalues and orthogonal eigenvectors. **This is the class PCA lives in.**
- **Diagonal:** zero off-diagonal. Acts as independent scaling along each axis.
- **Orthogonal:** $Q^T Q = Q Q^T = I$, so $Q^{-1} = Q^T$. Pure rotation/reflection; preserves lengths and angles. $\|Qx\| = \|x\|$. Eigenvalues all have $|\lambda| = 1$.
- **Positive semi-definite (PSD):** $x^T A x \geq 0$ for all x . Equivalent characterizations: all eigenvalues are ≥ 0 ; can be written as $B^T B$. Covariance matrices are PSD because $x^T \text{Cov}(X)x = \text{Var}(x^T X) \geq 0$. Valid kernel matrices are PSD: this is exactly Mercer's condition from SL6.
- **Positive definite (PD):** $x^T A x > 0$ for $x \neq 0$. All eigenvalues are > 0 . Has a unique Cholesky factorization $A = LL^T$.

2.3 Trace and Determinant: The Two Scalar Invariants

$$\text{tr}(A) = \sum_i A_{ii} = \sum_i \lambda_i, \quad \det(A) = \prod_i \lambda_i.$$

The trace is the sum of the diagonal and the sum of eigenvalues. The determinant is the signed volume scaling factor.

Properties used constantly:

- $\text{tr}(AB) = \text{tr}(BA)$ (cyclic), which appears in matrix-calculus identities.
- $\det(AB) = \det(A) \det(B)$.
- $\det(A) = 0 \iff A$ is singular $\iff$ some $\lambda_i = 0 \iff A$ is rank-deficient.

3 Eigenvectors and Eigenvalues: The Core Machinery

For a **square** matrix A , an **eigenvector** $v \neq 0$ is a direction that A only stretches, without rotating:

$$Av = \lambda v.$$

λ is the corresponding **eigenvalue**: the stretch factor along v .

3.1 How to Find Them

Rearrange:

$$(A - \lambda I)v = 0.$$

A non-zero solution v exists iff $(A - \lambda I)$ is singular, i.e.

$$\det(A - \lambda I) = 0 \quad (\text{characteristic equation}).$$

This is a polynomial of degree n in λ , the **characteristic polynomial**, so an $n \times n$ matrix has n eigenvalues, with multiplicity and possibly complex values.

Worked example. Let

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}.$$

Then

$$\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5).$$

So $\lambda_1 = 5$ and $\lambda_2 = 2$.

For $\lambda_1 = 5$, solve $(A - 5I)v = 0$:

$$\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Rightarrow \quad v_1 = v_2, \quad v^{(1)} = (1, 1).$$

For $\lambda_2 = 2$, solve $(A - 2I)v = 0$:

$$\begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Rightarrow \quad 2v_1 = -v_2, \quad v^{(2)} = (1, -2).$$

Notice $v^{(1)} \cdot v^{(2)} = 1 - 2 = -1 \neq 0$. These eigenvectors are **not orthogonal**, because A is **not symmetric**. Orthogonality only kicks in for symmetric A .

3.2 The Spectral Theorem: The Load-Bearing Fact for PCA

If A is real and symmetric, then:

- All n eigenvalues are real.
- There exists an orthonormal basis of eigenvectors.
- A can be written as $A = Q\Lambda Q^T$, where Q has eigenvectors as columns ($Q^T Q = I$) and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$.

The decomposition $A = Q\Lambda Q^T$ says: any symmetric transformation is “rotate into the eigenbasis” (Q^T), scale independently (Λ), and rotate back (Q). In the eigenbasis the matrix is diagonal; it acts on each axis independently.

3.3 Eigenvalue Identities You Will Actually Use

$$\text{tr}(A) = \sum_i \lambda_i, \quad \det(A) = \prod_i \lambda_i, \quad A^k v = \lambda^k v, \quad A^{-1} v = \frac{1}{\lambda} v \quad \text{if } A \text{ is invertible.}$$

The last two are why eigen-decomposition trivializes matrix powers and inverses: $A^k = Q\Lambda^k Q^T$, and Λ^k is just element-wise on the diagonal.

3.4 Geometric vs. Algebraic Multiplicity

- **Algebraic multiplicity** of λ : its multiplicity as a root of the characteristic polynomial.
- **Geometric multiplicity** of λ : $\dim \text{Null}(A - \lambda I)$, or how many linearly independent eigenvectors share that eigenvalue.

When these differ, A is **defective** and cannot be diagonalized. For symmetric matrices, the case we care about, they always agree.

4 PCA: The Eigenproblem of UL3, Worked End-to-End

Given a data matrix $X \in \mathbb{R}^{n \times d}$ with n samples and d features:

Step 1: Center.

Subtract the column mean: $\tilde{X} = X - \mathbf{1}\mu^\top$, where $\mu = \frac{1}{n}X^\top\mathbf{1}$. Without centering, the largest principal component just captures the offset from origin.

Step 2: Form the covariance matrix.

$$C = \frac{1}{n-1} \tilde{X}^\top \tilde{X} \in \mathbb{R}^{d \times d}.$$

C is symmetric and PSD by construction.

Step 3: Eigen-decompose.

$C = V\Lambda V^\top$. Sort eigenpairs by λ_i descending. Columns of V are the **principal components**, or orthonormal directions; the diagonal of Λ contains the **variances** captured along each direction.

Step 4: Project.

Keep top k PCs as V_k , and project: $Z = \tilde{X}V_k$. The projected data lives in $\mathbb{R}^k$, with $\text{Var}(Z_j) = \lambda_j$.

Step 5: Variance explained.

$$\text{Explained variance ratio of PC}_j = \frac{\lambda_j}{\sum_i \lambda_i}.$$

4.1 Worked Numerical Example

Tiny dataset, already centered:

$$\tilde{X} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ -2 & 0 \\ 0 & -1 \end{bmatrix}.$$

Covariance:

$$C = \frac{1}{3} \tilde{X}^\top \tilde{X} = \frac{1}{3} \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 8/3 & 0 \\ 0 & 2/3 \end{bmatrix}.$$

This is already diagonal, so the PCs are the standard axes: $v^{(1)} = (1, 0)$ with $\lambda_1 = 8/3$, and $v^{(2)} = (0, 1)$ with $\lambda_2 = 2/3$. PC1 captures

$$\frac{8/3}{8/3 + 2/3} = 80\%$$

of the variance, exactly what you would guess from the data's spread: four times wider along x than along y .

4.2 Why This Works: The One-Paragraph Proof Sketch

PCA's objective is to find a unit vector u maximizing the projected variance $\text{Var}(\tilde{X}u) = u^\top Cu$, subject to $\|u\| = 1$. The Lagrangian is

$$u^\top Cu - \lambda(u^\top u - 1),$$

and setting its gradient to zero yields $Cu = \lambda u$. So the optimal u is an **eigenvector of C** , and the maximum value of $u^\top Cu$ over unit u is the largest eigenvalue $\lambda_{\max}$. The next PC is found over the orthogonal complement, which gives the second-largest eigenvalue, and so on.

5 Singular Value Decomposition (SVD)

Eigen-decomposition needs a square matrix. **SVD generalizes it to any matrix $A \in \mathbb{R}^{m \times n}$:**

$$A = U\Sigma V^\top.$$

- $U \in \mathbb{R}^{m \times m}$ is orthogonal; columns are **left singular vectors**.
- $V \in \mathbb{R}^{n \times n}$ is orthogonal; columns are **right singular vectors**.
- $\Sigma \in \mathbb{R}^{m \times n}$ is diagonal-ish, with non-negative **singular values** $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$.

Interpretation: any linear map decomposes as rotation ($V^\top$), then axis-aligned scaling (Σ), then rotation (U).

5.1 Connection to Eigen-Decomposition

$$\begin{aligned} A^\top A &= V\Sigma^\top \Sigma V^\top = V(\Sigma^2)V^\top \Rightarrow \text{right singular vectors } V \text{ are eigenvectors of } A^\top A, \\ AA^\top &= U\Sigma\Sigma^\top U^\top = U(\Sigma^2)U^\top \Rightarrow \text{left singular vectors } U \text{ are eigenvectors of } AA^\top, \\ \sigma_i &= \sqrt{\text{eigenvalue}_i(A^\top A)}. \end{aligned}$$

So: SVD of $\tilde{X}$ directly produces PCA, with V equal to the PCs and $\sigma_i^2 = (n-1)\lambda_i$. In practice, **PCA is always computed via SVD on the data matrix**, never by forming the covariance matrix and eigen-decomposing it, because squaring condition numbers by forming $X^\top X$ destroys precision.

5.2 Low-Rank Approximation: Eckart–Young

Keep the top k singular values and vectors:

$$A_k = U_k \Sigma_k V_k^\top \quad (\text{sum of } k \text{ rank-1 outer products})$$

$$A_k = \sum_{j=1}^k \sigma_j u_j v_j^\top.$$

Eckart–Young theorem: A_k is the **best rank- k approximation** to A in both Frobenius and spectral norms:

$$\|A - A_k\|_F^2 = \sum_{j=k+1}^{\min(m,n)} \sigma_j^2.$$

This is the rigorous version of UL3's “minimize reconstruction error”: the error of dropping a component is exactly the sum of squared singular values you threw away.

5.3 Worked SVD Interpretation

For

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix},$$

already diagonal and already orthogonal-as-rotation-of-itself, the SVD is trivially $U = V = I$ and $\Sigma = \text{diag}(3, 1)$. The map sends the unit circle to an ellipse with semi-axes 3 along e_1 and 1 along e_2 . The singular values are the lengths of the ellipse's semi-axes when A is applied to the unit sphere, a useful mental picture in any dimension.

6 Matrix Calculus Identities

Notation: $f : \mathbb{R}^n \rightarrow \mathbb{R}$. The **gradient** $\nabla f \in \mathbb{R}^n$ is the column vector of partials. For $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the **Jacobian** $J_f \in \mathbb{R}^{m \times n}$ is $(J_f)_{ij} = \partial f_i / \partial x_j$.

Expression	Gradient with respect to x
$a^\top x$ ($= x^\top a$)	a
$x^\top A x$ (any A)	$(A + A^\top)x$; equals $2Ax$ when A is symmetric
$\ x\ ^2$	$2x$
$\ Ax - b\ ^2$	$2A^\top(Ax - b)$; least-squares normal equations
$\log \det(A)$ with respect to A	$A^{-\top}$

Worked check: derive the least-squares normal equations.

Minimize

$$L(\beta) = \|X\beta - y\|^2 = (X\beta - y)^\top (X\beta - y).$$

Expand:

$$L(\beta) = \beta^\top X^\top X \beta - 2\beta^\top X^\top y + y^\top y.$$

Take the gradient and set it to zero:

$$\nabla L = 2X^\top X \beta - 2X^\top y = 0 \quad \Rightarrow \quad X^\top X \beta = X^\top y \quad \Rightarrow \quad \hat{\beta} = (X^\top X)^{-1} X^\top y.$$

That is the closed-form OLS estimator, derived in three lines from the table above. Same trick for ridge: add $\lambda \|\beta\|^2$ and you get

$$\hat{\beta} = (X^\top X + \lambda I)^{-1} X^\top y.$$

7 Where It Shows Up in CS7641

Concept	Where
Inner products, projections	SL6 kernels/SVMs; kNN (SL4) and k-means (UL1) distances
PSD matrices	SL6 Mercer's condition; covariance matrices in PCA
Eigen-decomposition of covariance	UL3 PCA
SVD / low-rank approximation	UL3 PCA reconstruction; collaborative filtering
Orthogonality vs. independence	UL3 PCA (uncorrelated) vs. ICA (independent)
Linear maps, rank	UL3/UL4 dimensionality reduction in general
Matrix calculus	SL3 backprop; OL4 closed-form MLE/MAP
Lagrange multipliers on $u^\top C u$	PCA derivation itself; mirrors SL6's SVM

8 Review Questions with Sketched Answers

Conceptual Checks

1. **Geometrically, what does the dot product $x \cdot y$ measure?**

It measures alignment: $x \cdot y = \|x\| \|y\| \cos \theta$. Zero means orthogonal, which is what defines the SVM separating hyperplane ($w \cdot x + b = 0$) and what makes the kernel trick possible: the decision function uses data only via inner products.

2. **What is special about eigenvectors of a symmetric matrix?**

By the spectral theorem, they are **orthogonal** and the eigenvalues are **real**. Symmetric is the case that matters for PCA because covariance matrices are always symmetric, so principal components come out orthonormal for free.

3. **Difference between eigen-decomposition and SVD?**

Eigen-decomposition $A = Q\Lambda Q^{-1}$ requires A to be **square** and ideally diagonalizable. SVD $A = U\Sigma V^T$ works for **any** matrix, including rectangular data matrices. Eigen-decomposing a non-square data matrix is not defined; that is why PCA-via-SVD on the data matrix is the standard route.

Derivation / Math

4. **Eigenvector equation; eigenvalues of $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$.**

$Av = \lambda v$ and $\det(A - \lambda I) = (2 - \lambda)(3 - \lambda) = 0$. So $\lambda = 2, 3$ with eigenvectors $(1, 0)$ and $(0, 1)$, obvious because diagonal matrices act as independent scaling along the standard axes.

5. **Project x onto unit u .**

$\text{proj}_u(x) = (x \cdot u)u$. The scalar coefficient is $x \cdot u$; the projection's length is $|x \cdot u| = \|x\| |\cos \theta|$. Also, $x - (x \cdot u)u \perp u$.

6. **Why PCA equals eigen-decomposition of the covariance matrix.**

PCA maximizes $\text{Var}(\tilde{X}u) = u^T C u$ subject to $\|u\| = 1$. The Lagrangian is $u^T C u - \lambda(u^T u - 1)$. Gradient zero gives $Cu = \lambda u$. So the optimal directions are eigenvectors of C , and the variance captured along each is the corresponding eigenvalue.

Analysis: Why Does It Matter?

7. **PCA's components orthogonal vs. ICA's independent.**

Orthogonality is a linear-algebra property ($u^T v = 0$); independence is a statistical one, $p(x, y) = p(x)p(y)$. Independence implies uncorrelated, hence orthogonal in covariance space, but not the reverse. PCA's uncorrelated components can still be statistically dependent, such as in a $Y = X^2$ case. ICA goes further by minimizing mutual information.

8. **Why both covariance and kernel matrices must be PSD.**

PSD means $x^T A x \geq 0$ for all x . For a covariance matrix this is just $\text{Var}(x^T Z) \geq 0$; variance cannot be negative. For a kernel matrix it is Mercer's condition, guaranteeing the kernel implicitly corresponds to an inner product in some feature space. Without that, the SVM dual is meaningless.

9. **Why top- k SVD is the best rank- k approximation.**

Eckart–Young: the truncation error is exactly the sum of squared dropped singular values. Since SVD orders them largest-first, dropping the smallest minimizes total reconstruction error, formalizing UL3’s “minimize reconstruction error” objective.

Exam-Style

10. **Eigenvalues** [8, 5, 1, 0.5, 0.1]; **PCs to retain about 93%?**

Total = 14.6. Top 1: $8/14.6 \approx 55\%$. Top 2: $13/14.6 \approx 89\%$. Top 3: $14/14.6 \approx 96\%$. **Keep 3** to retain at least 93%.

11. **SVD as three operations.**

$A = U\Sigma V^\top$: rotate input space by $V^\top$, scale axes independently by Σ , then rotate output space by U . Any linear map is exactly this composition.

12. **Inner products in SL6, and the kernel trick.**

The dual form of the SVM contains data only as inner products $x_i \cdot x_j$. Replacing $x_i \cdot x_j$ with a kernel $k(x_i, x_j) = \phi(x_i) \cdot \phi(x_j)$ lets you implicitly operate in a high-dimensional ϕ -space without ever computing ϕ , provided k is PSD by Mercer’s condition.